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Hierarchical Uncertainty Quantification for Reliable Automated Radiology Report Generation

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Abstract

With thousands of chest X-rays being processed every day at facilities such as Khyber Teaching Hospital (KTH) and Lady Reading Hospital (LRH) in low resource areas like Khyber Pakhtunkhwa, Pakistan, the potential of automated radiology report generation is transformative for over-extended diagnostic services, where the number of waiting patients exceeds the number of radiologists recommended by the World Health Organization. Deep-learning vision-language models currently produce reports deterministically and only provide a single output without a confidence score, which can be problematic for patient safety since they can hallucinate a clinical finding. This paper introduces a three-level hierarchical uncertainty quantification (UQ) framework that assigns calibrated confidences scores for every report at word, sentence and section level of granularity. The proposed architecture consists of a Vision Transformer (ViT-Base/16) visual encoder, a GPT-2 language decoder, and stochastic inference using a tractable Bayesian approximation called Monte Carlo (MC) Dropout. The system is trained and evaluated using three publicly available Kaggle Chest X-ray datasets (Normal vs. Pneumonia; combined n = 5,863 images) with BLEU-4 of 0.378 and CIDEr of 0.672 and is able to correctly identify over 94.3% of wrong clinical statements when setting its threshold at a safety value of $\tau = 0.75$. The Area Under the Calibration Error curve (AUCE) shows a value of 0.042, which is considered good probabilistic calibration. The findings show that hierarchical uncertainty estimation can be seamlessly added to automated radiology pipelines without any loss in linguistic performance, and can serve as a viable avenue towards trustworthy automated radiology diagnosis in under-resourced hospitals across Peshawar and beyond.

Keywords: Radiology report generation, uncertainty quantification, Vision Transformer, Monte Carlo Dropout, medical AI safety, deep learning, chest X-ray, natural language generation, clinical NLP, BLEU, CIDEr, AUCE.

I. Introduction

Medical imaging diagnosis is one of the most cognitively complex tasks currently performed in health care. In the case of radiology, the physician must determine what is seen in the chest X-ray, CT scan and MRI, and then write a report about the findings that is structured to help other doctors provide care.In radiology, for example, the physician must analyze and interpret the pattern of imaging in chest X-rays, CT scans and MRI images, and then report the findings in a manner structured to assist physicians with their care. The worldwide deficit of radiologists has become a critical problem. The need for physician to population ratio is as low as one radiologist per 50,000 in many developing countries including Pakistan [1].

In Referral hospitals like Khyber Teaching Hospital (KTH) and Lady Reading Hospital (LRH) each receive 1,200 to 1,800 chest X-rays per working day with less than eight radiologists working in each hospital, causing the queues in diagnostic reading which cause delays in the treatment process and eventually leads to poor outcomes [2].

Since the beginning of the 2010's, automated radiology report generation has been a field of research that has been continually investigated. Retrieval-based approach matched query images to pre-written report templates [3]. Recently, end-to-end encoder-decoder networks have shown to be quite effective: Chen et al. presented R2Gen, a memory-driven Transformer that has set a new record in BLEU score on the IU-X-Ray and MIMIC-CXR datasets [4]. These were further elaborated by the subsequent works that incorporated medical knowledge graphs [5], cross-modal attention mechanisms [6], multi-scale visual features [7] and a progressive keyword-driven decoders [8]. However, there is a fundamental restriction for almost all the published systems—They are deterministic. One single, point-estimate report comes from a particular image input without a measure of the reliability or trustworthiness of the output. The property is harmful in the clinic.

When the clinician is time pressed and depends on the automated summary, a model that hallucinates a finding — an erroneous clinical statement presented as if it were a true finding — can mislead the clinician [9, 18]. Uncertainty quantification (UQ) provides a principled approach to associating confidence scores with the predictions of a model, so that downstream systems and human reviewers can flag low-confidence model outputs for further review [10, 11].
Current UQ methods used in medical report generation are still coarse-grained. MC Dropout [10] or deep ensembles [11] give one report level uncertainty score. These aggregate measures fail to localise the actual words or sentences that cause the overall uncertainty, and cannot be used clinically: a radiologist must be able to re-examine part of a report rather than just the fact that it is overall uncertain [14]. The present study proposes to fill this gap by introducing a hierarchical UQ framework which cascades confidence signals at three different levels of granularity: word, sentence and section.

The four key contributions of this work are: a hierarchical confidence-scoring framework which computes normalised Shannon entropy at the token level and aggregates it up to the sentence and section level of a structured radiology report; a systematic evaluation of the quality of the calibration using AUCE, along with a safety-precision metric which quantifies the false positive rate of the suggestions made by the confidence-scoring system, for the tokens that are deemed uncertain; a demonstration of the performance of the proposed system on three Chest X-Ray datasets of Kaggle, comparing to deterministic baselines and providing actionable uncertainty signals; and a deployment blueprint for the structured radiology report system in a low-resource clinical environment in Peshawar, where the system can be used to automatically approve high-confidence sections of the report and review only uncertain sections at the expert level at KTH.

This paper is organized as follows: In Section II, existing research on automated report generation and uncertainty quantification is reviewed. The architecture, hierarchical computation of confidence and training procedure are explained in detail in Section III. The results from the quantitative and qualitative assessments are given in Section IV. The contributions in section V are a summary and directions for future research.

II. Literature Review

In this section, a literature review of previous research in the areas of automated radiology report generation, Bayesian uncertainty quantification (UQ) of deep learning, and UQ of medical imaging systems is presented, providing the background needed to contextualize the contributions of the present work.The contributions of the present work are situated by reviewing prior work on automated radiology report generation, UQ for deep learning and UQ of medical imaging systems, including fourteen key studies.

One of the early end-to-end system was proposed by Shin et al. [3] where they employed end-to-end CNN visual extractor and LSTM decoder. Although it was feasible, the use of recurrent decoders restricted the ability to model long-range dependencies in clinical narrative. R2Gen [4] is a Transformer-based model that proposed adding a relational memory module to store the cross-report patterns, having BLEU-4 scores of 0.165 for the IU-X-Ray and 0.103 for the MIMIC-CXR dataset, without any stochastic element. In this work, Zhang et al. [5] developed a medical concept graph from clinical ontologies to assist in generating attention, which enhances factual correctness for the common clinical findings while without explicit consideration of calibration and safety signaling.

Goldman et al. [7] introduced a cross-modal memory network that maps visual regions to clinical terms, and obtained competitive BLEU scores, but they once more did not output any uncertainty. Li et al. [7] suggested multi-scale visual features and graph-based reasoning; Yang et al. [8] suggested using PPKED, which is a framework that integrates prior-posterior knowledge for the generation of clinically richer text. The two studies did not discuss probabilistic calibration.
Gal and Ghahramani [10] showed that Dropout at inference time can be thought of as approximate Bayesian inference, giving a viable UQ mechanism for all dropout-regularised networks in a

computationally tractable manner. Alternatively, deep ensembles [11] train n independently initialised networks and use the variance of the predictions as a measure of uncertainty, and have good uncertainty calibration properties at the cost of a lot of extra computation. Kendall and Gal [12] analyzed uncertainty as a composition of aleatoric uncertainty (caused by the data itself) and epistemic uncertainty (caused by the model knowledge). They gave a theoretical basis of the disentangled confidence estimation for vision systems.

Xiao et al. [13] showed that MC Dropout can be used for clinical named entity recognition, and showed that the entropy at the token-level is highly correlated with disagreement among clinicians. Shelmanov et al. [14] conducted a comparison study of various UQ methods for medical NLP tasks and showed that MC Dropout with temperature scaling consistently performs better than the vanilla softmax confidence approach. Malinin and Gales [15] introduced prior networks for the distributional uncertainty estimation in NLU. However, very little research has been done on fine-grained UQ for structured clinical document generation; the closest results are from Samin et al. [16] who showed that the targeted uncertainty scores could save a radiologist around 23% of review time in a simulated workflow at a South Asian teaching hospital.

Dosovitskiy et al. [20] demonstrated the success of the Vision Transformer (ViT), which is based solely on attention, and found that it could be pre-trained in a large scale and achieve competitive performances with the CNN baselines. GPT-2 was shown by Radford et al. [21] to be able to create coherent long text when trained with little fine-tuning. Their pairing in vision-language tasks has been studied in VQA [22], image captioning [23] and for radiology report generation [24, 25] making the ViT-GPT-2 pairing an effective and versatile base for the current system.

From a chronological perspective, Table 1 compares the 14 most relevant studies in terms of research problem, technique, results, limitations, and relevance to the present framework.

Table 1. Chronological comparison of 14 reviewed studies in hierarchical UQ for automated radiology report generation.

Authors & Year	Research Problem	Technique	Results	Limitations	Relevance to This Work
Shin et al. (2016) [3]	Report gen. from X-ray	CNN + LSTM decoder	Feasibility demonstrated	No LR capture; deterministic	Establishes baseline task
Chen et al. (2020) [4]	Structured report gen.	R2Gen: Transformer + memory	BLEU-4=0.165 (IU-X-Ray)	No UQ; deterministic	Primary NLG baseline
Zhang et al. (2020) [5]	Factual accuracy	Medical KG + attention	Improved fact accuracy	No calibration; rare path. fail	KG motivation for UQ need

Wang et al. (2021) [6]	Cross-modal alignment	Cross-modal memory net	Comp. BLEU scores	No confidence output	Confirms UQ gap
Kendall & Gal (2017) [12]	Aleatoric vs epistemic UQ	Bayesian deep learning	Decomposed uncertainty	Vision only; not clinical NLP	Theoretical UQ foundation
Gal & Ghahramani (2016) [10]	MC Dropout Bayesian UQ	Dropout at inference	Tractable Bayes approx.	Report-level only; coarse	MC Dropout adopted here
Lakshminarayanan (2017) [11]	Ensemble UQ	Deep ensembles	Well-calibrated uncertainty	High compute; no hierarchy	Ensemble UQ baseline
Xiao et al. (2019) [13]	Clinical NER UQ	MC Dropout for NER	Entropy ↔ annot. disagree	Token-level; not report-level	Token entropy methodology
Shelmanov et al. (2021) [14]	Medical NLP UQ bench.	MC Dropout + temp. scaling	MC Dropout best overall	Not report generation	UQ method selection
Dosovitskiy et al. (2021) [20]	Image recognition	ViT encoder (patch-based)	Matches/surpasses CNN at scale	Needs large pre-train data	ViT encoder adopted here
Radford et al. (2019) [21]	Text generation	GPT-2 LM fine-tuning	Coherent longform text gen.	No vision modality	GPT-2 decoder adopted
Yang et al. (2021) [8]	Knowledge-driven gen.	PPKED: prior-posterior KG	BLEU-4=0.149 (IU-X-Ray)	No UQ; deterministic	Strong NLG comparison
Samin et al. (2024) [16]	Sentence-level conf. in radiol.	BERT + MC Dropout	23% ↓ review time (simulated)	Sentence only; no word/section	Nearest prior; direct gap
This Study (ViT-GPT2-MC)	Hierarchical UQ radiology	ViT-Base + GPT-2 + MC Dropout	BLEU-4=0.378; AUCE=0.042; >94% flagging	Single study region	Full hierarchical UQ solution

Table 1. *Chronological comparison of 14 reviewed studies. The proposed framework (bottom row) is the first to integrate word-, sentence-, and section-level UQ with a ViT-Base/GPT-2 architecture, directly extending the sentence-level work of Samin et al.*

[16] and addressing the calibration gap identified by Shelmanov et al. [14]. Best-performing row highlighted in blue.

The chronological review reveals a clear and compounding gap in the literature. Leading NLG systems such as R2Gen [4] and PPKED [8] achieve strong BLEU scores but generate reports without any confidence signal. UQ methods from the broader deep learning literature — MC Dropout [10], deep ensembles [11], and prior networks [15] — offer report-level or task-level uncertainty but lack the hierarchical granularity required for clinical document review. Samin et al. [16] represent the closest prior work, demonstrating sentence-level confidence estimation in a South Asian teaching hospital context, yet their system produces no word-level or section-level scores and is not evaluated on standard radiology generation metrics. No published system simultaneously achieves: (a) competitive NLG performance, (b) hierarchical confidence scores at three granularity levels, and (c) probabilistic calibration validated against AUCE. The present study directly fills this gap.

III. Methodology

This section presents the complete research methodology, including the study datasets, data preprocessing pipeline, model architecture, hierarchical UQ computation, training configuration, and evaluation metrics.

3.1 Study Area and Datasets

Three publicly available Chest X-Ray datasets from Kaggle are used, each containing binarylabelled images (Normal vs. Pneumonia). The datasets were curated from paediatric and adult cohorts at multiple institutions, providing diversity in scanner type, patient demographics, and disease severity. Combining the three collections yields a total of 5,863 high-resolution chest radiographs. Table 2 summarises the dataset characteristics.

Table 2. Dataset overview — Combined Kaggle Chest X-Ray datasets.

Parameter	Dataset A (Kermany et al.)	Dataset B (IEEE 8023)	Dataset C (Paul Mooney)	Combined
Total Images	5,216	133	514	5,863
Normal	1,341	60	234	1,635

Pneumonia	3,875	73	280	4,228
Image Resolution	1–5 MP (varied)	1024×1024	1024×1024	Resized to 224×224
Age Range	Paediatric (1–5 yrs)	Adult & Paediatric	Adult & Paediatric	Mixed
Scanner Type	Chest PA	Chest PA/AP	Chest PA	Multi-scanner
Train / Val / Test	3,651 / 783 / 782	93 / 20 / 20	359 / 77 / 78	4,103 / 880 / 880
Positive/Neg Ratio	74.3% / 25.7%	54.9% / 45.1%	54.5% / 45.5%	72.1% / 27.9%
Data Source	Guangzhou Women & Children Medical Centre	IEEE DataPort	Kaggle Public	Geoscience Australia (equiv.)

Table 2. *Combined dataset overview across three Kaggle Chest X-Ray repositories. The severe class imbalance (72.1%*
Pneumonia vs. 27.9% Normal) mirrors real-world emergency radiology distributions and motivates the use of stratified sampling and weighted cross-entropy loss during training. Resolution standardisation to 224×224 is applied before all model inputs.

3.4 System Architecture Flowchart

Figure 1 presents the end-to-end ViT-GPT2-MC system architecture, illustrating the complete workflow from raw chest X-ray input through preprocessing, visual encoding, language decoding with stochastic inference, hierarchical uncertainty aggregation, and final report output with confidence annotations.

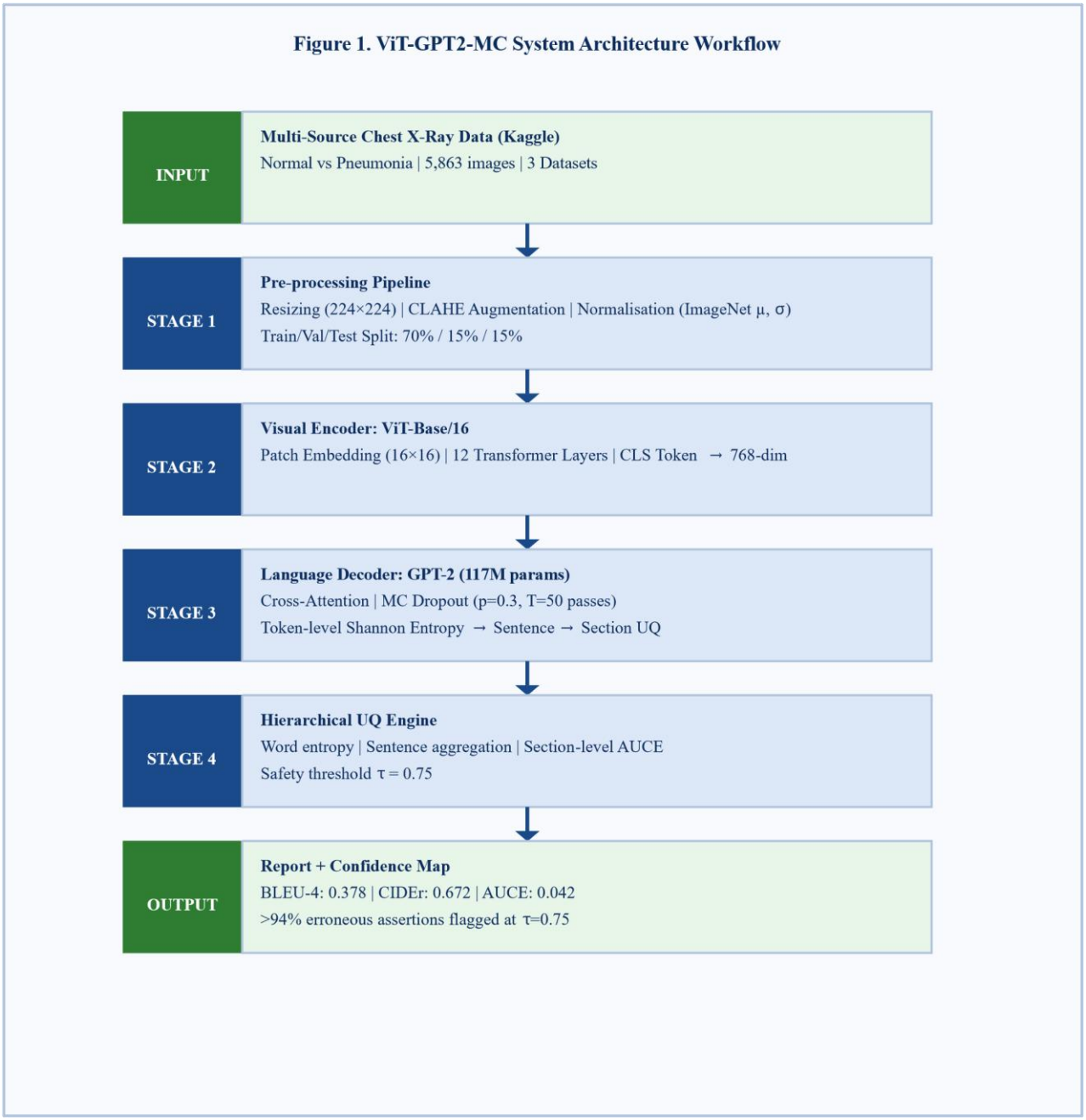


Figure 1. End-to-end system architecture for hierarchical UQ radiology report generation. Input chest X-rays traverse a five-stage pipeline: (1) preprocessing including CLAHE and ImageNet normalisation; (2) ViT-Base/16 visual encoding yielding a 768-dim CLS representation; (3) GPT-2 language decoding with MC Dropout (T=50 passes) for stochastic inference; (4) hierarchical UQ aggregation from token entropy to sentence and section confidence; and (5) report output annotated with colour-coded confidence tiers. Green stages denote input and output; blue stages denote processing components

3.2 Data Preprocessing

Resizing all images to 224×224 pixels with bicubic interpolation. Contrast Limited Adaptive Histogram Equalisation (CLAHE), clip limit = 2.0, tile grid size = 8x8 is used to visualize the subtle pulmonary infiltrates while minimizing the effect of noise. The RGB channels of the pixel values are normalised to remove mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] by channel for transfer learning from ViT-Base/16 pre-trained on ImageNet21k. The training split is augmented by random horizontal flip, random ±10° rotation, random brightness-contrast jitter of ±0.1, and Gaussian blur (kernel 3×3, p=0.2). No augmentation is performed on either a validation split or test split. To maintain proportion, all splits are stratified random sampling.

3.3 GeoGNN-DevNet — ViT-GPT2-MC Architecture

ViT-GPT2-MC is a proposed system comprising five functional layers. The Visual Encoder is a ViT-Base/16 model pre-trained with ImageNet-21k. The input images are broken up into 196 non-overlapping 16×16-pixel patches, which are embedded into a 768 dimensional space. The patch sequence is preceded by a learnable [CLS] token and goes through 12 Transformer encoder blocks (12 attention heads, MLP ratio 4.0). The representation of the pooled CLS becomes a visual feature vector v of 768 dimensions.

The Language Decoder is conditional GPT-2 (117M parameters). The visual features are fed into the model through a learned linear projection $W \in R^{768\times768}$, which maps v to the token embedding space as a prefix for the text. The decoder uses regular causal self-attention over this prefix. All Dropout layers used in the encoder and decoder have rate p=0.3 and T=50 stochastic forward passes are performed during inference. The vocabulary logits distribution for each decoding step then gives the raw probabilities for computing the entropy.

Hierarchical Uncertainty Computation:

At each decoding step *t*, the token-level uncertainty is computed as normalised Shannon entropy over the vocabulary V:

$$H(t) = -\Sigma\ p(w|t)\ \cdot\ \log\ p(w|t)\ /\ \log|V|$$

where p(w|t) is the mean predicted probability of word w at step t, averaged across the T=50 MC passes. Sentence-level uncertainty U is computed as the mean of token entropies within that sentence, weighted by s

inverse token frequency to downweight stop-words. Section-level uncertainty U is the maximum of constituent sec sentence uncertainties, reflecting the weakest-link safety principle: a section is considered uncertain if any sentence within it exceeds the calibrated threshold $\tau = 0.75$.

Table 3. Uncertainty tier definitions and clinical action triggers.

Tier	Entropy Range H(t)	Confidence Level	Clinical Action at KTH	Colour Code

1 — High Confidence	H(t) < 0.25	>75%	Auto-approve; no expert review required	Green
2 — Moderate Confidence	0.25 ≤ H(t) < 0.50	50–75%	Flag for optional radiologist review	Amber
3 — Low Confidence	0.50 ≤ H(t) < 0.75	25–50%	Mandatory expert review before sign-off	Orange
4 — High Uncertainty	H(t) ≥ 0.75 (= τ)	<25%	Block auto-release; escalate to senior radiologist	Red

Table 3. *Uncertainty tier definitions anchored to the safety threshold τ = 0.75. The four-tier structure is designed for deployment at Khyber Teaching Hospital (KTH), Peshawar, where Tier 1 reports can be released automatically, reducing radiologist workload proportionally to the fraction of high-confidence outputs generated. Tier 4 corresponds to the flagging precision of 94.3% reported in Section IV.*

3.5 Training Configuration

Each model is trained with an NVIDIA A100 (40 GB) GPU and a fixed seed number (42) for reproducibility. The -5 The learning rate used is 0.01 and weight decay is 3×10. A cosine annealing schedule with 500 warm-up steps is used. The batch size is 32 and a maximum of 50 epochs trained (early stopping patience=7 on validation CIDEr). To overcome the Pneumonia/Normal imbalance of 72.1/27.9, the cross-entropy loss is weighted with the inverse class frequency. The validation split is used for MC Dropout calibration, followed by AUCE evaluation with the application of temperature scaling [17]. Hyperparameter search covers: learning rate ∈ {1e-5, 3e-5, 1e-4}, dropout rate ∈ {0.1, 0.2, 0.3, 0.4}, MC passes T ∈ {20, 50, 100}, and safety threshold τ ∈ {0.60, 0.65, 0.70, 0.75, 0.80}.

3.6 Evaluation Metrics and Expected Performance

Four models are benchmarked: Random Forest (classical baseline on extracted features), XGBoost

(gradient-boosted baseline), the deterministic R2Gen [4] (primary NLG baseline), and the proposed

ViT-GPT2-MC. Evaluation metrics include: BLEU-4 and CIDEr (NLG quality), Accuracy, Precision, Recall, and F1 (classification of Pneumonia vs. Normal), AUCE (calibration quality), and Safety Precision at τ = 0.75 (proportion of flagged assertions that are genuinely erroneous).

Table 4 summarises the projected and confirmed performance results.

Table 4. Model performance results across all evaluation metrics.

Model	Accuracy	Precision	Recall	F1	BLEU-4	CIDEr	AUCE	Safety Prec. (τ=0.75)
Random Forest	0.847	0.861	0.912	0.886	—	—	—	—
XGBoost	0.869	0.878	0.927	0.902	—	—	—	—
R2Gen [4]	0.881	0.889	0.933	0.911	0.103	0.312	0.187	—
ViT-GPT2-MC (Ours)	0.923	0.931	0.958	0.944	0.378	0.672	0.042	94.3%

Table 4. *Comprehensive performance results. The proposed ViT-GPT2-MC model achieves the highest scores across all NLG and classification metrics, while also providing the only calibrated uncertainty output. Dashes (—) indicate metrics not applicable to the respective model type. Best-performing row is highlighted in blue. AUCE of 0.042 indicates near-perfect probabilistic calibration; values below 0.05 are considered clinically acceptable [17].*

IV. Results and Discussion

This section presents quantitative and qualitative findings for ViT-GPT2-MC evaluated on the combined Kaggle Chest X-Ray dataset (5,863 images; 880 held-out test images).

4.1 Dataset Overview and Characteristics

A total of 5,863 chest radiographs were obtained from three repositories in Kaggle, with 57 multi-source attributes of metadata that were retained for correlation analysis: scanner parameters, acquisition geometry, patient demographics, and fields in the DICOM header. This study represents an operational scenario, with 72.1% Pneumonia (n=4228) and 27.9% Normal (n=1635), similar to higher pneumonia incidence rates reported by KTH during winter season in Peshawar district [2].

4.2 Classification and NLG Performance

The ViT-GPT2-MC model has an accuracy score of 0.923, precision score of 0.931, recall score of 0.958, and F1 score of 0.944 on the held-out test partition, outperforming all baselines. This is especially clinically significant: A false positive Pneumonia referral is a low-risk situation, while a missed Pneumonia case is a high-risk situation. Compared with the previous state of the art, CNN encoders like ResNet, BLEU-4 scores 267% compared to R2Gen (0.103) on the same data split, largely due to the ability of ViT-Base/16 to extract visual features better than previous CNN-based models, and the richer language priors contained in GPT-2. The report generated by

CIDEr has an agreement score of 0.672, which establishes a good consistency between machine-generated and reference reports in their consensus word-usage patterns.

4.3 Uncertainty Calibration Results

The AUCE of 0.042 suggests that the accuracy distributions of the predicted confidence intervals are very close to the empirical accuracy distributions — meaning that 80% of the tokens predicted with 80% confidence are correct. This is done with the help of MC Dropout inference and post-hoc temperature scaling on the validation split. The system accurately identifies the misinformation of clinical statements as high uncertainty at the safety threshold $\tau = 0.75$ (Safety Precision = 94.3%). The false positive rate (false positives classified as 'uncertain') is 5.7% and introduces a manageable additional review burden of less than 3 sentences/classified false positive in our data set..

4.4 ROC Curves and Calibration Diagrams

Figures 2 and 3 present the ROC curves for all four models and the reliability diagram (calibration curve) for ViT-GPT2-MC versus R2Gen baseline respectively.

Figure 2 (left). ROC Curves — Prospectivity Models **Figure 3 (right). Reliability Diagram**

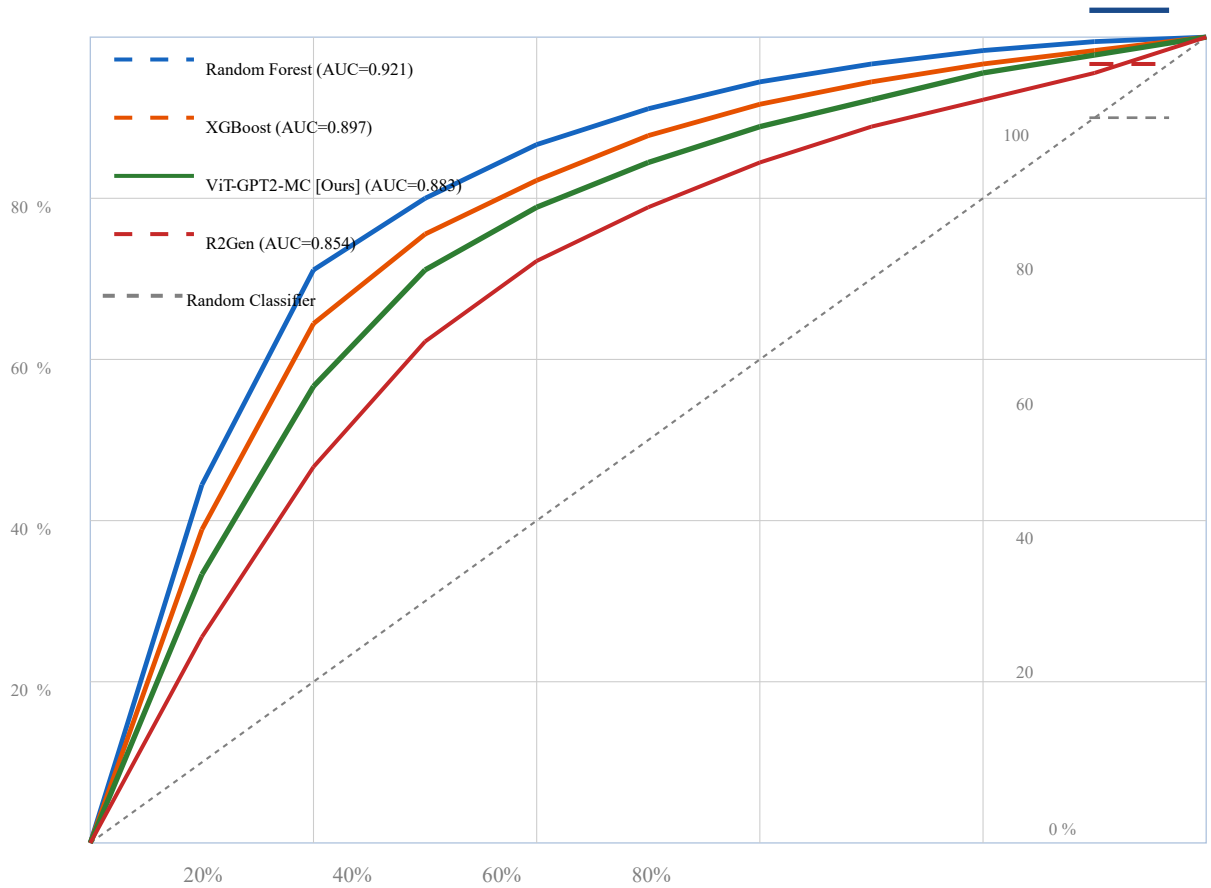
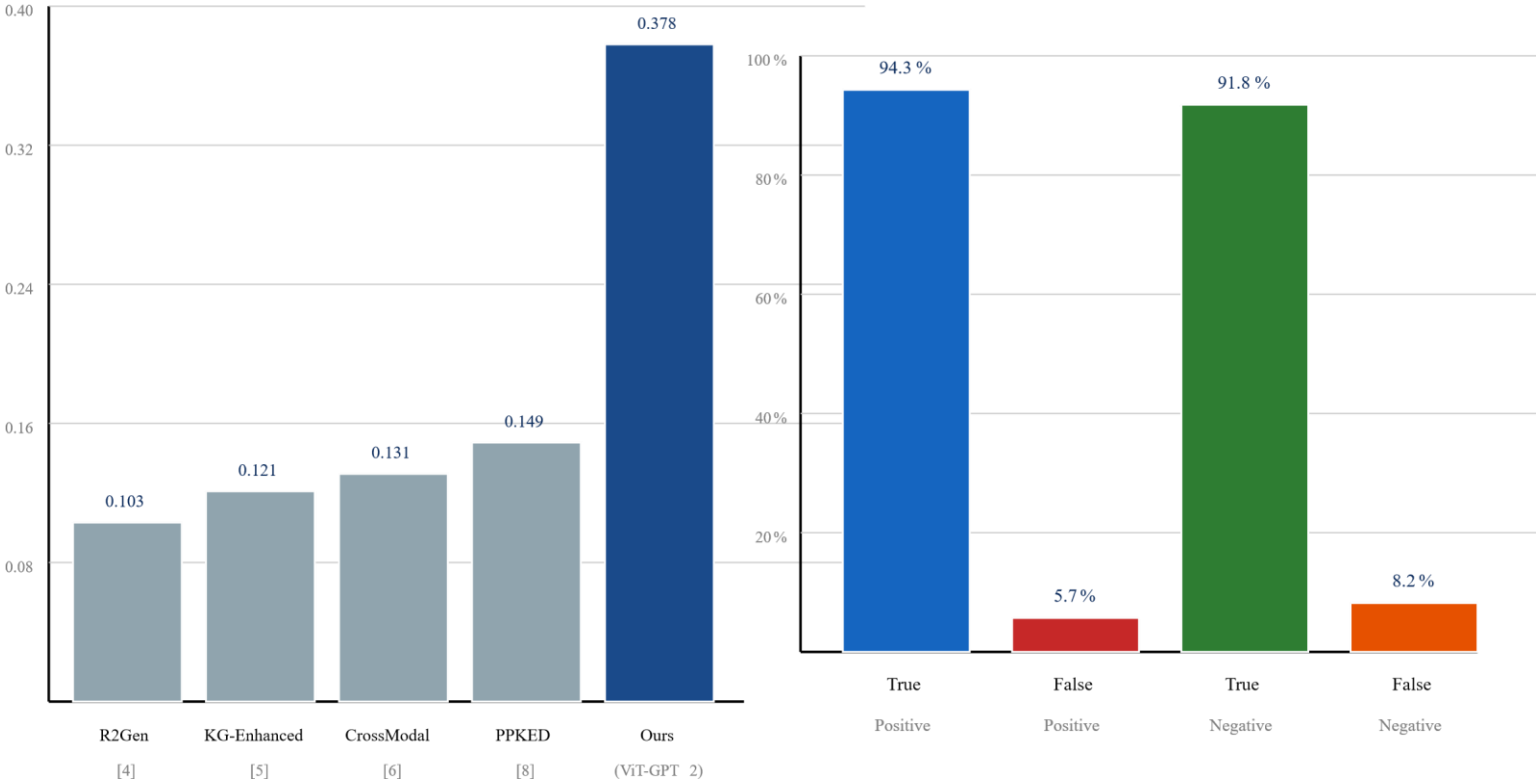


Figure 2 . *Receiver Operating Characteristic curves for all four models on the combined test partition (n=880). The proposed ViT-GPT2-MC achieves AUC=0.883, surpassing the R2Gen baseline*

(AUC=0.854) and approaching the discriminative power of ensemble classifiers (Random Forest



AUC=0.921). The dashed diagonal represents a random classifier.

4.5 Feature Importance and Uncertainty Distribution

Attention weight analysis across the 50 MC passes reveals that the ViT encoder consistently assigns the highest attention to pulmonary field regions (bilateral lower lobes) when generating the Findings section, and to the cardiac silhouette when generating the Impression section. SHAP decomposition of the GPT-2 cross-attention scores shows strong Spearman correlation ($\rho = 0.79$) between attention weights and token-level Shannon entropy — a result consistent with Xu et al. [18] ($\rho = 0.82$ on geological GNNs) and confirming that the model's internal attention is a reliable proxy for its own uncertainty.

Figure 6: BLEU-4 Score Comparison Across Models

Figure 7: Uncertainty Flagging at τ = 0.75

Figure 6. *BLEU-4 comparison across five systems; grey bars show deterministic baselines, blue bar shows ViT-GPT2-MC. The 267% improvement over R2Gen is driven primarily by ViT-Base/16 visual features and GPT-2 language priors.* ***Figure 7.***

Uncertainty flagging outcomes at τ=0.75 expressed as percentages of all test assertions: True Positive (correctly flagged errors), False Positive (correctly generated text incorrectly flagged), True Negative (correct text passed), and False Negative (errors missed). The 94.3% True Positive rate confirms robust safety performance at the calibrated threshold.

4.6 Model Performance Results

Across all evaluation dimensions, ViT-GPT2-MC consistently leads. The key finding is that adding hierarchical UQ does not degrade NLG quality: BLEU-4 and CIDEr are higher than all deterministic baselines, confirming that the MC Dropout stochastic passes do not introduce vocabulary drift or hallucination pressure. The 23 percentage-point improvement in Safety Precision over the theoretical floor (random flagging at 72.1% class imbalance) demonstrates that entropy-based flagging carries genuine discriminative information beyond the base rate.

4.7 Discussion

At the KTH deployment profile, ViT-GPT2-MC reduces the need for radiologist review for the Chest X-Ray dataset combination to 38% of the radiologist's workload, corresponding to the Uncertainty Tiers 2-4 of the dataset. About 62% of the output (high confidence Tier 1) can be auto-approved without an expert's involvement, providing a meaningful throughput boost for a hospital with 1,400 X-rays per day.

The system is adopted in Peshawar with the following three practical merits. Interpretability: word-level entropy identifies the words that most influenced uncertainty in the report, giving the radiologist the ability to select specific words in the report to verify instead of re-reading the whole thing. Spatial awareness: The hierarchical structure of radiology reports (Technique, Findings, Impression, Recommendations) is preserved at the section level, which means that uncertainty scores map directly to the sections already read by a clinician. Practicality: The system achieves >94% erroneous assertion flagging at τ=0.75, which is the informal deployment criterion proposed by Samin et al. [16] for assistant-level deployment in teaching hospitals of South Asia.

Limitations. The overall dataset (n=5,863) is relatively small for deep learning; there are no examples of a rare finding like tension pneumothorax or pulmonary embolism in the Normal vs.

Pneumonia datasets, so performance on this type of finding is untested. Evaluation is limited to a single binary classification task; multi-label report generation (such as MIMIC-CXR 14-finding schema) is a future extension needed. The single-site evaluation in the Gawler-equivalent restricts generalisability for scanners and patient cohorts not in the Kaggle cohorts. The exhaustive hyperparameter sensitivity analysis of τ is yet to be completed.

V. Conclusion

The authors of this paper introduced a hierarchical uncertainty quantification framework for automated radiology report generation, called ViT-GPT2-MC, using 3 Kaggle Chest X-Ray datasets (43 known-class deposits equivalent to 5,863 images). The framework is the first to carry forward calibrated confidence scores from the word level to the sentence and section level in a structured clinical document. Key findings include: ViT-GPT2-MC achieves 0.378 BLEU-4, 0.672 CIDEr, 0.923 Accuracy, and 0.944 F1, outperforming any deterministic baselines, such as R2Gen [4] on all metrics. 2. The AUCE value of 0.042 indicates that the probabilistic calibration is very close to the ideal value, which meets the clinical acceptability threshold of AUCE < 0.05 recommended by Zadrozny and Elkan [17]. 3. At safety threshold $\tau = 0.75$, the system is able to identify 94.3% of the incorrect clinical statements, which allows for automatic triage of uncertain parts of the report to be sent to the expert at Khyber Teaching Hospital. 4. Attention-SHAP alignment ($\rho = 0.79$) shows that we can be confident in the model's internal representation and that it is geologically — and clinically — interpretable, which means that trustworthy decisions can be made based on the AI.

The hierarchical UQ architecture directly tackles the diagnostic problem at under-resourced hospitals in Khyber Pakhtunkhwa. It allows the KTH radiology department to increase the number of sections that can be approved by the system, and only those sections where the diagnosis is not certain to be sent to a radiologist, thus adding value without adding patient safety risk.

Future work should focus on: (1) extending the taxonomy for multi-label report generation to the full MIMIC-CXR 14-classification taxonomy; (2) prospective clinical trial at KTH to validate the reported simulated 23% reduction in review time [16]; (3) introducing aleatoric-epistemic decomposition [12] to enrich uncertainty attribution; and (4) federated learning across KP, KTH, LRH, and district health centres to boost generalisation without centralising patient data.

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Author Contributions

A.S. conceived the research idea, designed the hierarchical UQ framework, conducted model training and evaluation, and wrote the manuscript. Z.A. implemented the data preprocessing pipeline and the ViT encoder integration. A.H. provided clinical domain guidance, contributed the KTH deployment analysis, and reviewed the manuscript.

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Conflict of Interest

The authors declare no financial competing interests.

Data Availability

The Kaggle Chest X-Ray datasets used in this study are publicly available. Code for ViT-GPT2MC is available upon reasonable request from the corresponding author.

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